

## 12 Functions

### 12.1 Functions

2. Suppose  $A = \{a, b, c, d\}$ ,  $B = \{2, 3, 4, 5, 6\}$  and

$$f = \{(a, 2), (b, 3), (c, 4), (d, 5)\}.$$

State the domain and range of  $f$ . Find  $f(b)$  and  $f(d)$ .

The domain of  $f$  is  $A$  and the range of  $f$  is  $\{2, 3, 4, 5\}$ .

$f(b) = 3$  and  $f(d) = 5$ .

4. There are eight different functions  $f : \{a, b, c\} \rightarrow \{0, 1\}$ . List them all. Diagrams will suffice.

$$f_1 = \{(a, 0), (b, 0), (c, 0)\}$$

$$f_2 = \{(a, 0), (b, 0), (c, 1)\}$$

$$f_3 = \{(a, 0), (b, 1), (c, 0)\}$$

$$f_4 = \{(a, 0), (b, 1), (c, 1)\}$$

$$f_5 = \{(a, 1), (b, 0), (c, 0)\}$$

$$f_6 = \{(a, 1), (b, 0), (c, 1)\}$$

$$f_7 = \{(a, 1), (b, 1), (c, 0)\}$$

$$f_8 = \{(a, 1), (b, 1), (c, 1)\}$$

6. Suppose  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  is defined as  $f = \{(x, 4x + 5) : x \in \mathbb{Z}\}$ . State the domain, codomain and range of  $f$ . Find  $f(10)$ .

The domain and codomain of  $f$  are both  $\mathbb{Z}$ . The range is the set of values  $4x + 5$  for  $x \in \mathbb{Z}$ , that is, the integers that are one more than a multiple of 4:

$$\{4x + 5 : x \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, 13, \dots\}.$$

**8. Consider the set  $f = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x + 3y = 4\}$ . Is this a function from  $\mathbb{Z}$  to  $\mathbb{Z}$ ? Explain.**

No. For  $f$  to be a function from  $\mathbb{Z}$  to  $\mathbb{Z}$ , every integer must occur as the first coordinate of some pair in  $f$ . But  $x = 0$  does not: that would require  $3y = 4$ , which has no integer solution  $y$ . So 0 is not the first coordinate of any element of  $f$ , and  $f$  is not a function from  $\mathbb{Z}$  to  $\mathbb{Z}$ .

**10. Consider the set  $f = \{(x^3, x) : x \in \mathbb{R}\}$ . Is this a function from  $\mathbb{R}$  to  $\mathbb{R}$ ? Explain.**

Yes, this is a function  $f : \mathbb{R} \rightarrow \mathbb{R}$ . Every real number  $k$  is the cube of exactly one real number, namely  $x = \sqrt[3]{k}$ , so each  $k \in \mathbb{R}$  occurs exactly once as the first coordinate of an ordered pair in  $f$ .

**12. Is the set  $\theta = \{((x, y), (3y, 2x, x + y)) : x, y \in \mathbb{R}\}$  a function? If so, what is its domain, codomain and range?**

Yes. Every first coordinate is a pair  $(x, y) \in \mathbb{R}^2$ , and each such pair occurs exactly once, so  $\theta$  is a function  $\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  given by  $\theta(x, y) = (3y, 2x, x + y)$ . Its domain is  $\mathbb{R}^2$  and its codomain is  $\mathbb{R}^3$ .

For the range, a triple  $(a, b, c) \in \mathbb{R}^3$  equals  $\theta(x, y)$  exactly when  $a = 3y$ ,  $b = 2x$  and  $c = x + y$ , that is, when  $y = \frac{a}{3}$ ,  $x = \frac{b}{2}$  and  $c = \frac{b}{2} + \frac{a}{3}$ . Conversely, given any  $(a, b, c)$  with  $c = \frac{a}{3} + \frac{b}{2}$ , taking  $x = \frac{b}{2}$  and  $y = \frac{a}{3}$  gives  $\theta(x, y) = (a, b, c)$ . Therefore the range is

$$\{(a, b, c) \in \mathbb{R}^3 : 2a + 3b = 6c\},$$

a plane through the origin in  $\mathbb{R}^3$ .

## 12.2 Injective and Surjective Functions

**2. Consider the logarithm function  $\ln : (0, \infty) \rightarrow \mathbb{R}$ . Decide whether this function is injective and whether it is surjective.**

This function is bijective. Observe that, by properties of the exponential function,  $e^{\ln x} = x$  for every  $x \in (0, \infty)$ , and  $\ln e^b = b$  for every  $b \in \mathbb{R}$ .

First we show  $\ln$  is injective. Suppose  $\ln m = \ln n$  for some  $m, n \in (0, \infty)$ . Applying the exponential function to both sides gives  $e^{\ln m} = e^{\ln n}$ , that is,  $m = n$ . Thus  $\ln$  is injective.

Next we show  $\ln$  is surjective. Take any  $b \in \mathbb{R}$ , and let  $a = e^b$ . Then  $a \in (0, \infty)$ , since  $e^b > 0$  for every real  $b$ , and  $\ln a = \ln e^b = b$ . So every  $b \in \mathbb{R}$  equals  $\ln a$  for some  $a$  in the domain, and  $\ln$  is surjective.

Therefore  $\ln$  is bijective since it is both injective and surjective on this domain.

**4. A function  $f : \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$  is defined as  $f(n) = (2n, n + 3)$ . Verify whether this function is injective and whether it is surjective.**

This function is injective but not surjective.

First we show  $f$  is injective. Suppose  $f(m) = f(n)$  for some  $m, n \in \mathbb{Z}$ . This means  $(2m, m + 3) = (2n, n + 3)$ , thus  $2m = 2n$ . Dividing this equation by two gives  $m = n$ . Thus  $f$  is injective.

Next we show  $f$  is not surjective. Consider  $(1, 0) \in \mathbb{Z} \times \mathbb{Z}$ . If  $f(n) = (1, 0)$  for some  $n \in \mathbb{Z}$ , then equating the first coordinates gives  $2n = 1$ , which has no solution  $n \in \mathbb{Z}$ . So no integer maps to  $(1, 0)$ , and  $f$  is not surjective.

**6. A function  $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$  is defined as  $f(m, n) = 3n - 4m$ . Verify whether this function is injective and whether it is surjective.**

This function is surjective but not injective.

First we show  $f$  is not injective. Observe that  $f(0, 0) = 3(0) - 4(0) = 0$  and  $f(3, 4) = 3(4) - 4(3) = 12 - 12 = 0$ . Thus  $f(0, 0) = f(3, 4)$  even though  $(0, 0) \neq (3, 4)$ , so  $f$  is not injective.

Next we show  $f$  is surjective. Take any  $b \in \mathbb{Z}$ . Then  $(-b, -b) \in \mathbb{Z} \times \mathbb{Z}$ , and  $f(-b, -b) = 3(-b) - 4(-b) = -3b + 4b = b$ . So every  $b \in \mathbb{Z}$  equals  $f(m, n)$  for some  $(m, n)$  in the domain, and  $f$  is surjective.

**8. A function  $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$  is defined as  $f(m, n) = (m + n, 2m + n)$ . Verify whether this function is injective and whether it is surjective.**

This function is both injective and surjective, hence bijective.

First we show  $f$  is injective. Suppose  $f(m, n) = f(k, \ell)$  for some  $(m, n), (k, \ell) \in \mathbb{Z} \times \mathbb{Z}$ . This means  $(m + n, 2m + n) = (k + \ell, 2k + \ell)$ . Equating corresponding coordinates gives the two equations  $m + n = k + \ell$  and  $2m + n = 2k + \ell$ . Subtracting the first from the second yields  $m = k$ , and substituting this back into the first gives  $n = \ell$ . Therefore  $(m, n) = (k, \ell)$ , so  $f$  is injective.

Next we show  $f$  is surjective. Take any  $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ , and let  $m = b - a$  and  $n = 2a - b$ . So  $(m, n) \in \mathbb{Z} \times \mathbb{Z}$ , and  $f(m, n) = ((b - a) + (2a - b), 2(b - a) + (2a - b)) = (a, b)$ . So every  $(a, b) \in \mathbb{Z} \times \mathbb{Z}$  equals  $f(m, n)$  for some  $(m, n)$  in the domain, and  $f$  is surjective.

Being both injective and surjective,  $f$  is bijective.

**10. Prove the function  $f : \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{1\}$  defined by  $f(x) = \left(\frac{x+1}{x-1}\right)^3$  is bijective.**

*Proof.* First we show  $f$  is injective. Suppose  $f(x) = f(y)$  for some  $x, y \in \mathbb{R} - \{1\}$ , that is,

$$\left(\frac{x+1}{x-1}\right)^3 = \left(\frac{y+1}{y-1}\right)^3.$$

Every real number has exactly one real cube root, so taking cube roots of both sides gives

$$\frac{x+1}{x-1} = \frac{y+1}{y-1}.$$

Since  $x \neq 1$  and  $y \neq 1$ , both denominators are nonzero, so we may cross-multiply to get  $(x+1)(y-1) = (y+1)(x-1)$ . Expanding both sides,

$$xy - x + y - 1 = xy - y + x - 1.$$

Cancelling  $xy$  and  $-1$  leaves  $-x + y = -y + x$ , that is,  $2y = 2x$ , so  $x = y$ . Thus  $f$  is injective.

Next we show  $f$  is surjective. Take any  $b \in \mathbb{R} - \{1\}$  and let  $c = \sqrt[3]{b}$ . Note that  $c \neq 1$ , for otherwise  $b = c^3 = 1$ . Now let

$$x = \frac{c+1}{c-1},$$

which is defined because  $c \neq 1$ . Also  $x \neq 1$ , since  $x = 1$  would give  $c+1 = c-1$ , which is impossible. Hence  $x \in \mathbb{R} - \{1\}$ . Observe that

$$x+1 = \frac{c+1}{c-1} + 1 = \frac{2c}{c-1} \quad \text{and} \quad x-1 = \frac{c+1}{c-1} - 1 = \frac{2}{c-1},$$

so  $\frac{x+1}{x-1} = \frac{2c}{2} = c$ , and therefore

$$f(x) = \left(\frac{x+1}{x-1}\right)^3 = c^3 = b.$$

So every  $b \in \mathbb{R} - \{1\}$  equals  $f(x)$  for some  $x$  in the domain, and  $f$  is surjective.

Being both injective and surjective,  $f$  is bijective. □

**12. Consider the function  $\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}$  defined as  $\theta(a, b) = a - 2ab + b$ . Is  $\theta$  injective? Is it surjective? Bijective? Explain.**

This function is bijective. Since the first coordinate  $a$  can only be 0 or 1, we compute the two cases separately:  $\theta(0, b) = 0 - 2(0)b + b = b$  and  $\theta(1, b) = 1 - 2b + b = 1 - b$ . As  $b$  ranges over  $\mathbb{N}$ , the first line produces exactly the integers  $1, 2, 3, \dots$ , and the second produces exactly  $0, -1, -2, \dots$ , that is, the integers less than or equal to 0.

**14. Consider the function  $\theta : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$  defined as  $\theta(X) = \overline{\overline{X}}$ . Is  $\theta$  injective? Is it surjective? Bijective? Explain.**

This function is a bijection, to show this, note that  $\overline{\overline{\overline{X}}} = X$  for every  $X \subseteq \mathbb{Z}$ .

To see that  $\theta$  is injective, suppose  $\theta(X) = \theta(Y)$  for some  $X, Y \in \mathcal{P}(\mathbb{Z})$ , that is,  $\overline{X} = \overline{Y}$ . Taking complements of both sides gives  $\overline{\overline{X}} = \overline{\overline{Y}}$ , that is,  $X = Y$ . Thus  $\theta$  is bijective.

To see that  $\theta$  is surjective, take any  $Y \in \mathcal{P}(\mathbb{Z})$ . Then  $\overline{Y}$  is also a subset of  $\mathbb{Z}$ , so  $\overline{Y} \in \mathcal{P}(\mathbb{Z})$ , and  $\theta(\overline{Y}) = \overline{\overline{Y}} = Y$ .

Being both injective and surjective,  $\theta$  is bijective.

16. This question concerns functions  $f : \{A, B, C, D, E\} \rightarrow \{1, 2, 3, 4, 5, 6, 7\}$ . How many such functions are there? How many of these functions are injective? How many are surjective? How many are bijective?

### 12.3 The Pigeonhole Principle

2. Prove that if  $a$  is a natural number, then there exist two unequal natural numbers  $k$  and  $\ell$  for which  $a^k - a^\ell$  is divisible by 10.

4. Consider a square whose side-length is one unit. Select any five points from inside this square. Prove that at least two of these points are within  $\frac{\sqrt{2}}{2}$  units of each other.

6. Given a sphere  $S$ , a *great circle* of  $S$  is the intersection of  $S$  with a plane through its center. Every great circle divides  $S$  into two parts. A *hemisphere* is the union of the great circle and one of these two parts. Prove that if five points are placed arbitrarily on  $S$ , then there is a hemisphere that contains four of them.

### 12.4 Composition

2. Suppose  $A = \{1, 2, 3, 4\}$ ,  $B = \{0, 1, 2\}$ ,  $C = \{1, 2, 3\}$ . Let  $f : A \rightarrow B$  be

$$f = \{(1, 0), (2, 1), (3, 2), (4, 0)\},$$

and  $g : B \rightarrow C$  be  $g = \{(0, 1), (1, 1), (2, 3)\}$ . Find  $g \circ f$ .

4. Suppose  $A = \{a, b, c\}$ . Let  $f : A \rightarrow A$  be the function

$$f = \{(a, c), (b, c), (c, c)\},$$

and let  $g : A \rightarrow A$  be the function  $g = \{(a, a), (b, b), (c, a)\}$ . Find  $g \circ f$  and  $f \circ g$ .

6. Consider the functions  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  defined as  $f(x) = \frac{1}{x^2 + 1}$  and  $g(x) = 3x + 2$ . Find the formulas for  $g \circ f$  and  $f \circ g$ .

8. Consider the functions  $f, g : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$  defined as  $f(m, n) = (3m - 4n, 2m + n)$  and  $g(m, n) = (5m + n, m)$ . Find the formulas for  $g \circ f$  and  $f \circ g$ .

10. Consider the function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by the formula  $f(x, y) = (xy, x^3)$ . Find a formula for  $f \circ f$ .

## 12.5 Inverse Functions

2. In Exercise 9 of Section 12.2 you proved that  $f : \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{5\}$  defined by  $f(x) = \frac{5x + 1}{x - 2}$  is bijective. Now find its inverse.

4. The function  $f : \mathbb{R} \rightarrow (0, \infty)$  defined as  $f(x) = e^{x^3 + 1}$  is bijective. Find its inverse.

6. The function  $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$  defined by the formula  $f(m, n) = (5m + 4n, 4m + 3n)$  is bijective. Find its inverse.

8. Is the function  $\theta : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$  defined as  $\theta(X) = \overline{X}$  bijective? If so, what is its inverse?

10. Consider  $f : \mathbb{N} \rightarrow \mathbb{Z}$  defined as  $f(n) = \frac{(-1)^n(2n - 1) + 1}{4}$ . This function is bijective by Exercise 18 in Section 12.2. Find its inverse.

## 12.6 Image and Preimage

2. Consider the function  $f : \{1, 2, 3, 4, 5, 6, 7\} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$  given as

$$f = \{(1, 3), (2, 8), (3, 3), (4, 1), (5, 2), (6, 4), (7, 6)\}.$$

Find:

$$f(\{1, 2, 3\}), \quad f(\{4, 5, 6, 7\}), \quad f(\emptyset), \quad f^{-1}(\{0, 5, 9\}) \quad \text{and} \quad f^{-1}(\{0, 3, 5, 9\}).$$

4. This problem concerns functions

$$f : \{1, 2, 3, 4, 5, 6, 7, 8\} \rightarrow \{0, 1, 2, 3, 4, 5, 6\}.$$

How many such functions have the property that  $|f^{-1}(\{2\})| = 4$ ?

6. Given a function  $f : A \rightarrow B$  and a subset  $Y \subseteq B$ , is  $f(f^{-1}(Y)) = Y$  always true? Prove or give a counterexample.

8. Given a function  $f : A \rightarrow B$  and subsets  $W, X \subseteq A$ , then  $f(W \cap X) = f(W) \cap f(X)$  is false in general. Produce a counterexample.

10. Given  $f : A \rightarrow B$  and subsets  $Y, Z \subseteq B$ , prove  $f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z)$ .

12. Consider  $f : A \rightarrow B$ . Prove that  $f$  is injective if and only if  $X = f^{-1}(f(X))$  for all  $X \subseteq A$ . Prove that  $f$  is surjective if and only if  $f(f^{-1}(Y)) = Y$  for all  $Y \subseteq B$ .

14. Let  $f : A \rightarrow B$  be a function, and  $Y \subseteq B$ . Prove or disprove:  $f^{-1}(f(f^{-1}(Y))) = f^{-1}(Y)$ .